Step 4 **Establish a plan for resolving the problem**—You decide to plug in the network cable. This is a very simple resolution that does not affect other users. In other situations, your plan might involve informing coworkers of what is about to happen or possibly filing a request for formal change management.

Step 5 **Implement the solution or escalate the problem**—You plug in the cable.

Step 6 **Verify functionality and implement preventive measures**—You open your browser and find you can surf the web. You verify company network resources are available from File Explorer.

Step 7 **Document findings, actions, and outcomes**—This simple problem and solution don't require formal documentation. However, network technicians are generally expected to document troubleshooting tasks and solutions. In this case, you simply inform your coworkers that your network connection is now working.

Self-Check

15. What should you do after making a plan to solve a problem and carrying out your plan?
    a. Document your work.
    b. Find out what has recently changed.
    c. Test your theory.
    d. Verify functionality.
16. While exploring the problem and developing a theory about the cause, where can you go to get information on similar problems your coworkers have faced in the past?
    a. Safety data sheet
    b. Knowledge base
    c. Troubleshooting ticket in the ticket tracking system
    d. Instruction manual

☐ Check your answers at the end of this chapter.

Summary

Section 1-1: Network Models

- The term physical topology refers to a network's hardware and how devices and cables fit together. Logical topology refers to the way software controls access to network resources and how those resources are shared on the network.
- A peer-to-peer network model allows each computer's operating system to control access to its resources without centralized control. P2P networks are simple to configure and relatively inexpensive. But they are not scalable, not necessarily secure, and not practical for managing more than just a few computers.
- In a client–server network model, the NOS manages resources via a centralized directory database. Common examples of NOSs include Windows Server, Ubuntu Server, and Red Hat Enterprise Linux (Ubuntu and Red Hat are versions of Linux).

Section 1-2: Network Services

- Resources a network makes available to its users include applications and data, collectively referred to as network services.
- Networked devices use methods and rules for communication known as protocols. The two primary network protocols are TCP (Transmission Control Protocol) and IP (Internet Protocol), and the suite of all the protocols an OS uses for communication on a network is the TCP/IP suite of protocols.
- The primary protocol used by web servers and browsers (clients) is HTTP (Hypertext Transfer Protocol). When HTTP is layered on top of an encryption protocol, such as SSL (Secure Sockets Layer) or TLS (Transport Layer Security), the result is HTTPS (HTTP Secure), which secures the transmission.

- An email client uses SMTP (Simple Mail Transfer Protocol) to send an email message to the first server, which is an SMTP server. The first server sends the message on to the receiver's mail server, which delivers the message to the receiving client using one of two protocols: POP3 (Post Office Protocol, version 3) or IMAP4 (Internet Message Access Protocol, version 4).
- A DBMS (database management system) is software that makes requested changes to data and organizes data for viewing, reporting, or exporting. Many DBMSs use the programming language SQL (Structured Query Language) to configure and interact with the database's objects and data. Popular examples of SQL database software include Microsoft SQL Server, Oracle Database, and the open-source MySQL.

Section 1-3: Network Hardware

- In a LAN (local area network), each node on the network can communicate directly with others on the network. LANs are usually contained in a small space, such as an office or building. A switch connects LAN devices; it receives incoming data from one of its ports and redirects (or switches) it to another port or multiple ports that will send the data to its intended destination(s).
- In a star topology, all devices connect to one central device. In a mesh topology, each device connects to multiple other devices.
- A network port can be an onboard network port embedded in the computer's motherboard. Another type of port is provided by a NIC (network interface card), also called a network adapter, installed in an expansion slot on the motherboard.
- A router is a device that manages traffic between two or more networks and can help find the best path for traffic to get from one network to another. In SOHO (small office–home office) networks, which typically have fewer than 10 computers, a consumer-grade router is used to connect the LAN to the Internet. An enterprise or industrial-grade router can have several network ports, one for each of the networks it connects to.
- The fundamental difference between a switch and a router is that a switch belongs only to its local network and a router can belong to two or more networks.
- *Networked* devices connect to the network to access or provide resources, such as a file server, smartphone, smart thermostat, security camera, or network printer. A *networking* device is a device that enables connections on the network but does not, itself, provide applications or data for networked devices to access.
- A group of LANs spread over a wide geographical area is called a WAN (wide area network). A group of connected LANs in the same geographical area is known as a MAN (metropolitan area network) or CAN (campus area network), and these terms might be used interchangeably. WANs and MANs often use different transmission methods and media than LANs do. A much smaller network type is a PAN (personal area network), which is a network of personal devices, such as the connection between a smartphone and computer.

Section 1-4: The Seven-Layer OSI Model

- The seven-layer OSI (Open Systems Interconnection) reference model was developed to categorize the layers of network communication. Mapping each protocol onto the OSI model helps identify the logistics of which software program or device is initiating and/or receiving the communication and how other protocols are relating to it.
- The application layer in the OSI model describes the interface between two applications, each on separate computers. Protocols used at this layer include HTTP, SMTP, POP3, IMAP4, FTP, Telnet, and RDP.
- The presentation layer is responsible for reformatting, compressing, and/or encrypting data in a way that the application on the receiving end can read. The session layer of the OSI model describes how data between applications is synced and recovered if messages don't arrive intact at the receiving application. The application, presentation, and session layers are so intertwined that, in practice, it's often difficult to distinguish between them.
- The transport layer is responsible for transporting application layer payloads from one application to another. The two main transport layer protocols are TCP, which guarantees delivery, and UDP, which does not. Transport-layer protocols add control information in an area at the beginning of the payload called the header to create a message ready to be handed down to the network layer. The process of adding a header to the data inherited from the layer above is called encapsulation.
- The network layer is responsible for moving messages from one node to another until they reach the destination host. The principal protocol used by the network layer is IP (Internet Protocol). IP adds its own network layer header to the segment or datagram, and the entire network layer message is now called a packet. The network layer header identifies the sending and receiving hosts by their IP addresses.
- Layer 2 (the data link layer) and layer 1 (the physical layer) are responsible for interfacing with the physical hardware on the local network. Examples of data link layer protocols are Ethernet and Wi-Fi. The data link layer puts its own control information in a data link layer header that also

contains the hardware addresses of the source and destination NICs; it then attaches control information to the end of the packet in a trailer. The entire data link layer message is called a frame. The physical layer sends the frame's bits via a wired or wireless transmission.

Section 1-5: Safety Procedures and Policies

- A fire suppression system is designed to combat the outbreak of a fire and might include an emergency alert system, portable fire extinguishers, an emergency power-off switch, and a suppression agent such as a foaming chemical, gas, or water.
- A fail-open, or fail-safe, policy is often based on common sense to ensure that, in an emergency, no one is harmed when a system is not working. A fail-close, or fail-secure, policy is usually based on the need for security to protect private data or other resources even if that means access is denied until the emergency is over.
- An SDS (safety data sheet), formerly called an MSDS (material safety data sheet), explains how to properly handle substances such as chemical solvents, how to respond to accidental exposure, and how to dispose of them.
- Electrical and tool safety in workplaces is generally regulated by OSHA (Occupational Safety and Health Administration), which is the main federal agency charged with safety and health in the workplace. OSHA regulations for electrical safety require that electrical devices be turned off and the electrical supply locked out before employees work near these devices.
- Back injury, caused by lifting heavy objects, is one of the most common injuries that happens at work. Don't try to lift an object that is too heavy for you. Because there are no exact guidelines for how heavy is too heavy, use your best judgment as to when to ask for help.
- Sensitive electronic components can be damaged by ESD (electrostatic discharge), commonly known as static electricity. A discharge of up to 1,500 volts of ESD won't produce a visible spark or human-detectable shock. However, it only takes 10 volts to damage some components. Wear an ESD strap clipped to the computer chassis to eliminate ESD between you and the chassis and its components. Always shut down and unplug a computer before working inside it.

Section 1-6: Troubleshooting Network Problems

- As you gather information about a problem, identify the symptoms, question the user, find out what has recently changed, and determine the scope of the problem.
- Consider multiple approaches to identify a problem's cause, such as working through the layers of the OSI model. Troubleshooters generally follow the bottom-to-top OSI model by first suspecting and eliminating failed hardware as the cause. Some situations are obviously software related, and it makes more sense to follow the OSI layers top-to-bottom.
- Test your theory to assure yourself that it will indeed solve the problem before you implement the solution, and then determine the next steps needed. If your test proves your theory is wrong, move on to another guess or escalate the problem.
- Before you implement a fix, consider the scope of your change, especially how it will affect users, their applications, and their data. Check with your organization's policies to determine whether the proposed change falls under change management requirements.
- Before you make the change, be sure to alert all affected users in advance, create backups of software and data as needed, and save or write down current settings before you change them. Test your solution thoroughly, and clean up after yourself when you're finished. For complex problems, you might need to escalate the problem to someone with access to more technical resources or with more authority to test or implement a solution.
- Test the system for full system functionality. Also consider what you can do to make sure the problem doesn't reappear.
- Document problems and their resolutions, lessons learned, and unique or insightful solutions to problems in your company's call tracking system and knowledge base.

Key Terms

For definitions of key terms, check the Glossary.

AD (Active Directory)
AD DS (Active Directory Domain Service)
API (application programming interface)
application layer
backbone
bus topology
change management
client
client-server network model
connectionless protocol
connection-oriented protocol
data link layer
datagram

DBMS (database management system)
decapsulation
DNS (Domain Name System)
domain
encapsulation
ESD (electrostatic discharge)
fail close
fail open
fire suppression system
firewall
firmware
fragmentation
frame
FTP (File Transfer Protocol)
FTPS (FTP Secure or FTP over SSL)
grounding
header
host
HTTP (Hypertext Transfer Protocol)
HTTPS (HTTP Secure)
hub-and-spoke topology
hybrid topology
IMAP4 (Internet Message Access Protocol, version 4)
IP (Internet Protocol)
IP address
ISP (Internet service provider)
knowledge base
LAN (local area network)
logical topology
MAC (Media Access Control) address
mesh topology
network
network layer
network service
NIC (network interface card)
node
NOS (network operating system)
open source
OSHA (Occupational Safety and Health Administration)
OSI (Open Systems Interconnection) reference model
P2P (peer-to-peer) network model
packet
PAN (personal area network)
password manager
payload
physical layer
physical topology
POP3 (Post Office Protocol, version 3)
port
PPE (personal protective equipment)
presentation layer
protocol
RDP (Remote Desktop Protocol)
router
SAN (storage area network)
scalable
SDS (safety data sheet)
segment
server
server operating system
session layer
SFTP (Secure File Transfer Protocol)
SMTP (Simple Mail Transfer Protocol)
SNMP (Simple Network Management Protocol)
SOHO (small office-home office) network
SQL (Structured Query Language)
SSH (Secure Shell)
SSL (Secure Sockets Layer)
star topology
static electricity
switch
TCP (Transmission Control Protocol)
TCP/IP suite
Telnet
TLS (Transport Layer Security)
topology
trailer
transport layer
UDP (User Datagram Protocol)
WAN (wide area network)

Acronyms Checklist

The acronyms in Table 1-3 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

Table 1-3 Chapter 1 acronyms

Acronym	Spelled out
API	application programming interface
ARP	Address Resolution Protocol
CLI	command-line interface
FTP	File Transfer Protocol
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICMP	Internet Control Message Protocol
IP	Internet Protocol
LAN	local area network
MAC	Media Access Control

Continues

Acronym	Spelled out
NIC	network interface card
OS	operating system
OSI	Open Systems Interconnection
RDP	Remote Desktop Protocol
SAN	storage area network
SMTP	Simple Mail Transfer Protocol
SNMP	Simple Network Management Protocol

Acronym	Spelled out
SQL	Structured Query Language
SSH	Secure Shell
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
WAN	wide area network

Review Questions

1. In the client–server model, what is the primary secure protocol used for communication between a browser and web server?
   a. HTTPS
   b. TLS
   c. HTTP
   d. SSL
2. A network consists of five computers, all running Windows Professional. All the computers are connected to a switch, which is connected to a router, which is connected to the Internet. Which logical networking model does the network use?
   a. Hub-and-spoke
   b. Ring
   c. Hybrid
   d. Peer-to-peer
3. A network consists of seven computers and a network printer, all connected directly to one switch. Which network topology does this network use?
   a. Client-server
   b. Mesh
   c. Hub-and-spoke
   d. Star
4. Which two encryption protocols might be used to secure transmissions for email services?
   a. HTTP and HTTPS
   b. SSL and TLS
   c. FTP and SFTP
   d. SSH and RDP
5. What is the fundamental distinction between a layer 2 switch and a router?
   a. Devices connect to a switch using a hybrid topology, while devices connect to a router using a star topology.
   b. A switch has one NIC, and a router has many NICs.
   c. A switch is a combo device with many built-in functionalities, while a router only connects networks to each other.
   d. A switch belongs to a single LAN, while a router belongs to multiple LANs.
6. Which of the following is an application layer protocol?
   a. IP
   b. RDP
   c. TCP
   d. Apache
7. Which protocol functions at the transport layer of the OSI model?
   a. TLS
   b. IP
   c. UDP
   d. Wi-Fi
8. At the network layer, what type of address is used to identify the receiving host?
   a. MAC address
   b. Hardware address
   c. Port number
   d. IP address

9. As you're working to fix a problem with an application, you make multiple changes at once hoping that something will solve the issues you're having. You end up with more problems than when you started. Which step, if followed correctly, would have prevented this complication?
   a. Identify the problem.
   b. Test the theory to determine the cause.
   c. Establish a plan of action to resolve the problem and identify potential effects.
   d. Document findings, actions, outcomes, and lessons learned.

10. In the event of a fire, a failure policy that ensures all doors remain unlocked is called a _____ policy.
   a. power-off
   b. fail-close
   c. fail-open
   d. shutdown

Scenario-Based Questions

Question 1-1

A computer is unable to access the network. When you check the LED lights near the computer's network port, you discover the lights are not lit. Which layer of the OSI model are you using to troubleshoot this problem?

a. Physical layer
b. Network layer
c. Transport layer
d. Application layer

Question 1-2

A user complains that their computer cannot access email, although the computer can access websites. At which layer of the OSI model should you begin troubleshooting this problem and why?

a. Transport layer
b. Physical layer
c. Application layer
d. Data link layer

Question 1-3

While troubleshooting a problem, you realize the problem is caused by a complex series of issues that will affect a large number of users even to test your theory as to the cause, and the testing process won't even solve the problem. What should you do next in the troubleshooting process?

a. Test your theory to determine the cause.
b. Establish a plan for resolving the problem.
c. Establish a theory of probable cause.
d. Escalate the problem.

Hands-On Projects

Note 1-20 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided in Appendix D for evaluating student performance on these projects.

Project 1-1: IT and Networking Certifications

Estimated time: 20 minutes (+10 minutes for group work, if assigned)
Objective 1.6: Compare and contrast network topologies, architectures, and types.
Group work: This project includes enhancements when assigned as a group project.
Resources:
- Internet access

Description:
This course prepares you to take the CompTIA Network+ N10-009 exam, which is considered a fundamental benchmark toward a career in IT. Many other IT certifications will also promote success in your IT, networking, or cybersecurity career.

Use the web to research and respond to the following questions:

1. Which certification does CompTIA recommend a candidate for the CompTIA Network+ certification to already have before taking this exam? Include the web address of your authoritative source along with your response.

Caution | When you provide a source for your responses, you should always ensure the source is authoritative for the information you are giving. Don't rely solely on hearsay, informal discussion on discussion forums, or secondhand knowledge in blogs, wikis, or articles. For example, an authoritative source about a CompTIA exam would be a CompTIA-published webpage.

2. How long does CompTIA recommend you work in networking before you take the CompTIA Network+ exam? Include the web address of your authoritative source along with your response.
3. Cisco offers a full range of certifications focused on all aspects of networking. How long does Cisco recommend you work in networking before you take the most current CCNA exam for certification? Include the web address of your authoritative source along with your response.
4. Microsoft network-related certifications have shifted focus toward their cloud-based Azure platform. The entry-level Azure certification is called Microsoft Azure Fundamentals. Which Azure architectural components and services should Azure Fundamentals certification candidates be familiar with before taking the exam? Include the web address of your authoritative source along with your response.
5. AWS (Amazon Web Services) offers extensive certification options in various areas of cloud computing expertise. The entry-level AWS exam is the AWS Certified Cloud Practitioner certification. How long and in what roles does AWS recommend you work with the AWS cloud before you take the Cloud Practitioner exam? Include the web address of your authoritative source along with your response.

6. Search online for a job opening in IT networking in your geographical area that requires or recommends a degree, specific IT skills, and at least one IT certification. **Take a screenshot** of the job description and requirements; submit this visual with your responses to this project's questions. (Excellent sites that post IT jobs are **Indeed.com** and **Monster.com**.)
7. Respond to the following questions about the job listed:
   a. Which degrees are required or recommended?
   b. What types of skills are required or recommended?
   c. What work experience is required or recommended?
   d. Which IT certifications are required or recommended?
   e. What is the advertised salary range?
8. **For group assignments:** Each member of the group should research online for practice questions for one of the IT certifications discussed in this project. Group members should compare their sources to ensure no one uses the same source as someone else. Each group member quizzes the other group members with the practice questions and tallies their performance. Each group member then lists which exam objectives the group collectively demonstrates sufficient knowledge of as required by each objective covered by that exam. Submit the name of the exam, the source of the practice questions, a brief summary of the group's performance on the practice questions, and the list of exam objectives currently mastered by the group collectively.

Project 1-2: Create a Password Manager Account

Estimated time: 20 minutes
Objective 3.5: Compare and contrast network access and management methods.
Resources:
- Internet access
- (Optional) Personal smartphone or tablet

Description:
Throughout this course, you will create several accounts at different websites to access tools for various projects. As you read in the chapter, a password manager can help you document those passwords and store them securely. This course uses the open-source Bitwarden password manager to store account credentials from projects.

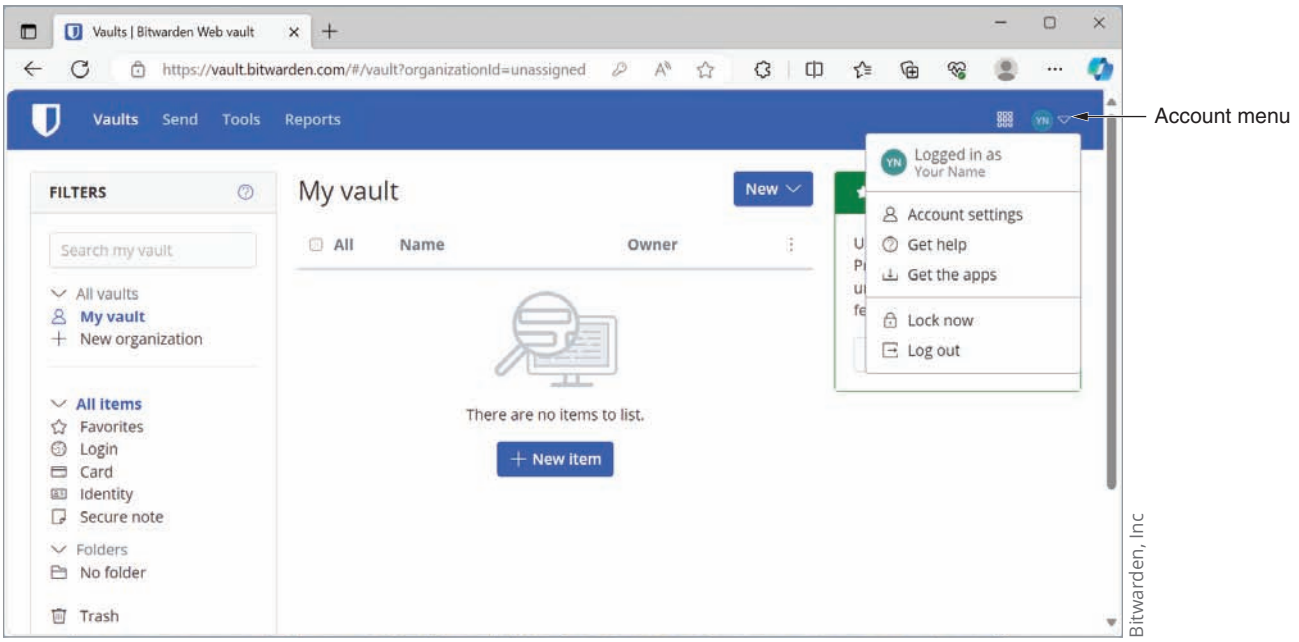
In this project, you create a Bitwarden account where you can store all your account information for the projects in this course. Bitwarden provides a free subscription option, and you can access your information from any device where you have signed into your account. If you want, you can also store account information for your other school and personal accounts in Bitwarden. Just remember to always keep your master password secure.

Note 1-21 | You'll learn more about how to create a secure password later. For now, keep in mind that the longer the password, the better. A simple and memorable way to do this is to think of an obscure line from a favorite song or movie, and use several words of that line to create your master password along with some extra numbers or symbols to add complexity. Here's an example: Spock$%Transmit@#Now.

Complete the following steps to create your password manager account:

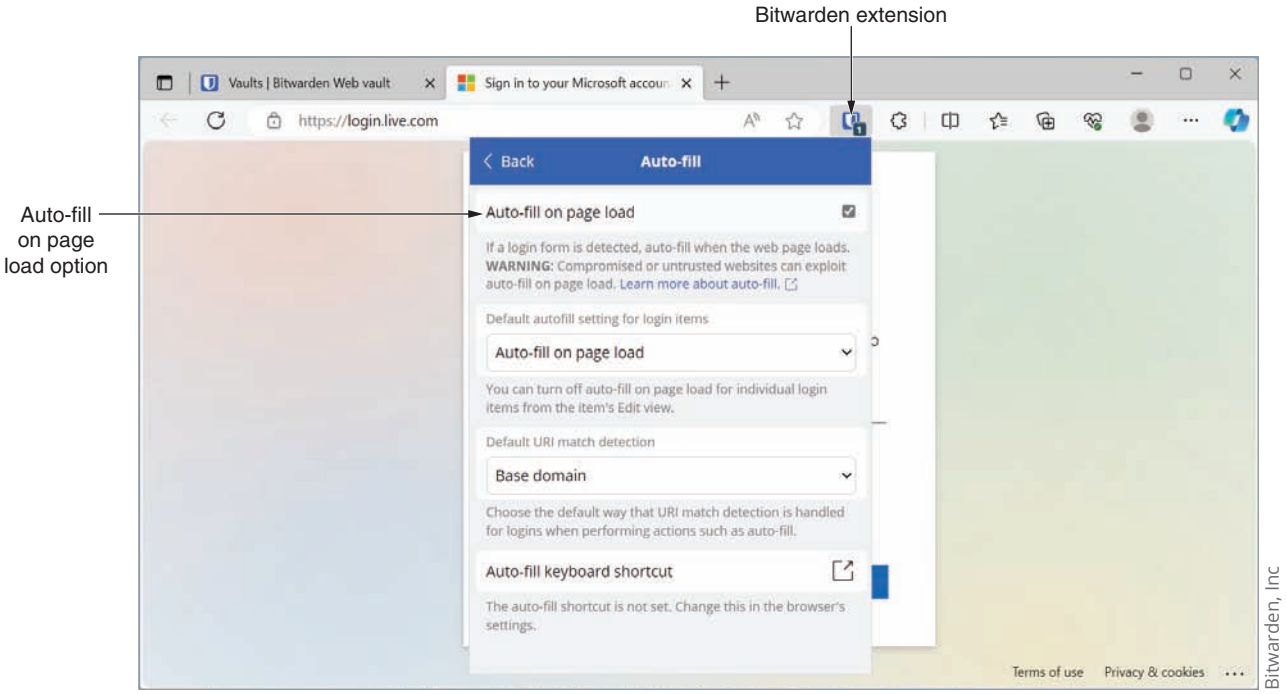
1. Go to **bitwarden.com**, click **Personal**, and click **Get started today**.
2. Create an account, making sure to create a secure master password (the longer, the better–just make sure you can remember it or store it somewhere else safely). Confirm your master password and give yourself a reminder if it's helpful.
3. In your browser, sign into your account, and make sure to choose the free plan, when given the option.
4. Verify your email address.
5. If you're working on your own computer or mobile device, you can install the Bitwarden app and/or the web browser extension, and then log into it. Whether on the website, in the app, or in the browser extension, click through to get to your Bitwarden vault. **Take a screenshot** of your empty vault; submit this visual with your responses to this project's questions. If you already have a Bitwarden account that you will be using for this course, be sure to obscure any private information from your screenshot. Note that no one, not even your instructor, will need access to your Bitwarden account for this course.
6. Click through the pages and menus to explore the vault (Figure 1-24). Respond to the following questions:
   a. What two-step login options are available with Bitwarden?
   b. On which page can you change your vault timeout setting?
   c. What is the purpose of the fingerprint phrase?

Figure 1-24 Store passwords, notes, and other secrets in your Bitwarden vault



7. Click the **New item** button in your vault. Enter information for a site you visit often, such as a social media site or an email service. If you prefer, you can make up information for this entry and then delete it later.
8. Launch the site from Bitwarden, which opens the site in another tab. Make any adjustments to ensure that Bitwarden will automatically open and sign into that site. For example, when using the Edge browser extension, click the Bitwarden extension, click **Auto-fill settings**, and then select the **Auto-fill on page load** check box (Figure 1-25).

Figure 1-25 Enable auto-fill



9. Log out of Bitwarden in your browser. Always remember to log out of your account before walking away from your computer. Store a copy of your master password in a very secure place, such as a lockbox in your home, a safe deposit box at a bank, or an encrypted file on your computer.

Note 1-22 | You can download and install Bitwarden as an app on your Windows, macOS, or Linux computer, or as an extension in your favorite browser on each computer that you own. Bitwarden is compatible with Chrome, Firefox, Safari, Edge, and other browsers. You can also install the Bitwarden app on your smartphone (Android or iPhone).

Caution | No password manager is 100 percent reliably secure. Hackers target these services, and occasionally they're successful. Bitwarden, however, is one of the most reliable password managers currently available for free. KeePass, 1Password, Keeper, LastPass, and Dashlane are also popular password managers. If you prefer, you can use one of these other password managers for this course.

No password manager is secure if you leave your account open on a computer or device you're not using, or if you write your master password where someone else can find it. Follow these guidelines consistently:

- Never save passwords in your browser, only in a password manager account such as the Bitwarden browser extension.
- Always log out of your password manager account when you're not using it.
- Always close browser windows where you have been signed into a secure account of any kind.
- Always lock or sign out of your OS before walking away from your computer.

Project 1-3: Apply Troubleshooting Methodology

Estimated time: 20 minutes
Objective 5.1: Explain the troubleshooting methodology.
Group work: This project includes enhancements when assigned as a group project.
Resources:

- A drawing app, such as Paint in Windows or a web app such as jspaint.app, kleki.com, or app.diagrams.net

Description:
Most likely at this point in your IT career, you've already encountered some challenging troubleshooting scenarios with computers, mobile devices, and perhaps even with networks. Interestingly, you probably intuitively applied some sound troubleshooting principles to the problem-solving process, and you might even have incorporated a basic understanding of networking layers as you worked through to a solution.

Complete the following steps:

1. Think back to one of the more interesting scenarios you've faced, one where you were able to solve the problem. Take a few moments to write down the symptoms you encountered, the information you gathered, and the questions you asked. Try to remember the sense of confusion or concern that this unknowing created.
2. Think through what theories you developed on the possible causes of the problem as well as what attempts you made to solve the problem. Write down as many details as you can remember about how you finally discovered the solution, and how you arrived at that conclusion.
3. Review the troubleshooting flowchart in Figure 1-23. Using a drawing app such as Microsoft Paint in Windows or a free web app such as jspaint.app, kleki.com, or app.diagrams.net, map your problem-solving experience to the steps illustrated in the flowchart and include additional details as they come to you. **Save this image** as a .png file; submit this visual with your responses to this project's questions.

After developing your troubleshooting diagram, respond to the following questions:

4. What do you notice about your progression through the OSI model layers? Even without necessarily knowing what the OSI model is, did you naturally take a top-to-bottom or a bottom-to-top approach to the problem?
5. What theories did you test that turned out to be wrong? What information or insights did you learn from those dead ends?
6. Did you involve anyone else in the problem-solving process? If so, who else was involved, and how did they help?
7. What did you do to test your solution? What measures did you take to ensure the problem didn't happen again?
8. Considering what you've now learned about troubleshooting methodology, what could you have reasonably done differently to discover the solution more quickly?
9. **For group assignments:** Each member of the group should write a summary of the problem experienced in their scenario, steps taken, and outcome of the issue as if they were documenting this information in a knowledge base. Next, exchange this documentation with another member of the group. Each member then reads through the information written by their classmate and lists questions they still have about the events or information gaps that could cause problems in the future. Discuss your concerns with the author of the scenario. Submit this information and a summary of the group discussion.

Capstone Projects

Note 1-23 | The Capstone Projects in this course are designed to give you a "big picture" experience of networking. While the Capstones don't always map closely to a chapter's learning objectives, each Capstone adds resources and skills to your toolkit while building on Capstones you completed in earlier chapters. For example, in this chapter, you will create a VM (virtual machine) in a hypervisor. In later Capstones, you will revisit this VM to complete other tasks on it. For those Capstones, you won't start from scratch—you'll build on resources you worked on in earlier chapters. Over time, your stream of learning will build momentum so that you will accomplish complex tasks in later Capstones that you could not have completed in a single, isolated project.

Sometimes, the Capstones will introduce you to concepts you haven't yet learned much about. Don't worry—detailed steps will guide you through each project. Then, when you study these concepts in later chapters, you'll already have some familiarity with the concepts. For example, you haven't yet learned about virtualization in this course, and yet, you're working with a hypervisor in this chapter's Capstone Project. By the time you get to the virtualization chapter, you'll have a decent understanding of what a hypervisor is and basically how it works, and that chapter's material will make a lot more sense to you.

Overall, these Capstones are intended to challenge you and also to provide a fun opportunity to apply what you're learning over time—to link concepts from chapter to chapter. In the study of learning science, this technique is called interleaving. Take good notes as you go, think creatively about what you're doing in each Capstone, and look for the ways each chapter's Capstones connect to other chapters. Enjoy!

In Capstone Project 1-1, you set up a VM (virtual machine) using Oracle VirtualBox. You will continue to build your virtual network of VMs in later chapters. VirtualBox is a client hypervisor, which is software used to manage VMs installed on a workstation. If you don't want to use VirtualBox as your hypervisor of choice, you can substitute another client hypervisor, such as Windows Hyper-V (included in Windows 10/11 Pro, Enterprise, and Education) or VMware Workstation Player (can be downloaded free from vmware.com). Most of the future VM projects in this course can be completed using any of these three hypervisors, and the instructions are written specifically for VirtualBox.

Note that Windows Hyper-V and Oracle VirtualBox don't play well on the same computer and can cause problems, such as failed network connectivity, if used concurrently. For that reason, don't install Hyper-V and VirtualBox on the same computer.

Note 1-24 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided in Appendix D for evaluating student performance on these projects.

Capstone Project 1-1: Set Up a Windows Virtual Machine Using VirtualBox

Estimated time: 45 minutes

Objective 1.2: Compare and contrast networking appliances, applications, and functions.

Resources:

- Any edition of Windows 10 or 11 installed on a 64-bit computer that supports UEFI.
  - Project instructions are written for Windows 11 hosts. However, Oracle VirtualBox can also be installed on a current or recent version of Windows, Linux, macOS, or Solaris.
  - While some editions of Windows can be installed on a 32-bit host, later Capstone VMs will require the host to be 64-bit, and all Capstone VMs are expected to be installed on the same host.
- Windows ISO file to install in a guest VM.

Description:

In this project, you download and install Oracle VirtualBox, which is a free hypervisor, to create and manage VMs and virtual networks on a single workstation. You'll first enable the workstation UEFI to support virtualization. You'll then create a VM in VirtualBox and install a Windows OS on the VM.

Note 1-25 | If you are using Hyper-V instead of VirtualBox for your hypervisor, be aware there are separate steps required to enable Hyper-V instead of installing it. Those steps are not covered here, and you will need to do some additional research online to help you enable Hyper-V.

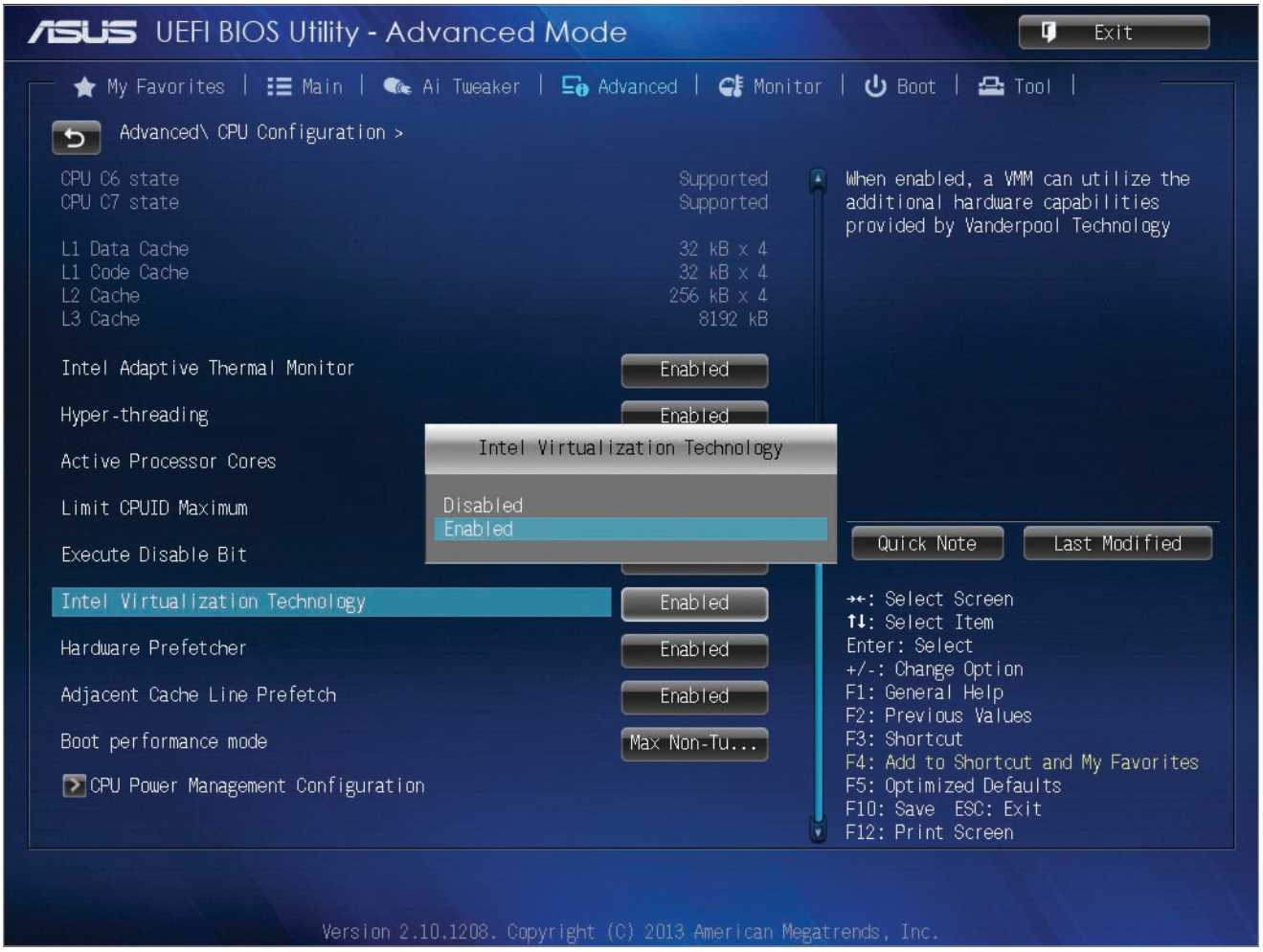
Complete the following steps to enable virtualization in UEFI and install VirtualBox:

1. If you are using a 64-bit host computer and want to install a 64-bit OS in the VM, HAV (hardware-assisted virtualization) must be enabled in UEFI setup. If you are not sure it is enabled, click **Start** and **Power**. Hold down the **Shift** key and click **Restart**. When the computer reboots, click **Troubleshoot**, **Advanced options**, and **UEFI Firmware Settings**. The computer reboots again, this time into UEFI setup.

Note 1-26 | Some motherboards might not show "UEFI Firmware Settings" as an option on the Advanced options screen. If this is the case for you, you'll need to do a little experimenting and troubleshooting. First, determine your motherboard manufacturer and model. To do this, continue the boot to Windows, press **Win+R**, and enter **msinfo32**, which will list the motherboard manufacturer and model on the System Summary page. Find the motherboard's documentation online to ensure it supports UEFI. If it does, you can try entering the UEFI settings during boot by pressing the required key, such as Esc, Del, F2, F4, F8, or F12. (If you're not sure which key to try, check your motherboard documentation or watch for a message during boot.) Before pressing the key to successfully access UEFI, you might first need to disable fast startup in the Windows Control Panel's Power Options menu. If you have trouble with any of this, be sure to search online for the problem you're having and look for information to help you figure it out. Learning how to research a problem online is an important skill for any IT technician.

2. Make sure hardware-assisted virtualization (HAV) is enabled. For the system shown in Figure 1-26, that's done on the CPU Configuration screen. Also make sure that any subcategory items under HAV are enabled. Save your changes, exit UEFI setup, and allow the system to restart to Windows.

Figure 1-26 Virtualization must be enabled in UEFI setup



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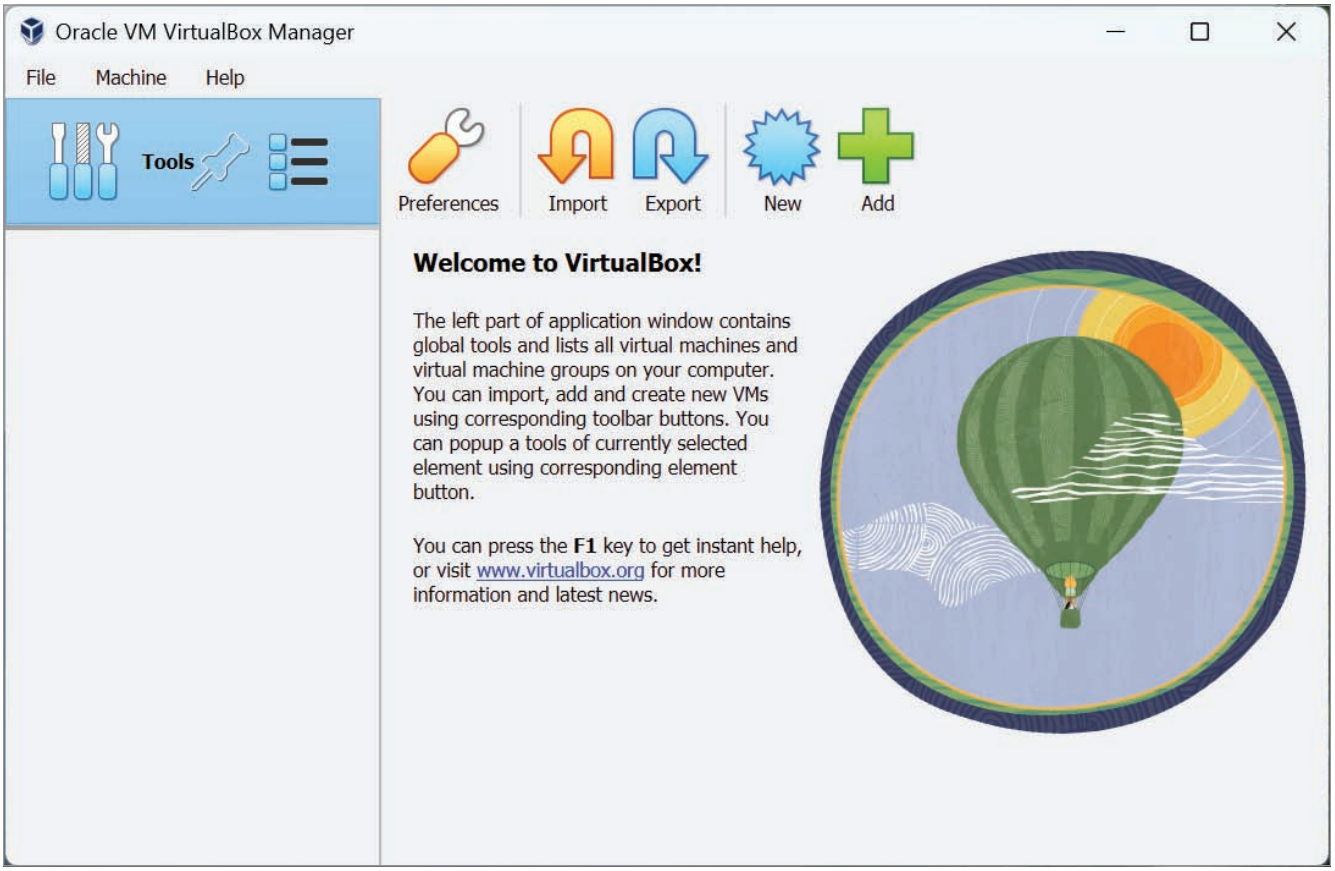
3. Go to **virtualbox.org/wiki/Downloads** and download the most current **VirtualBox platform package** for Windows hosts to your desktop or other folder on your hard drive. Install the software, accepting default settings during the installation. The Oracle VM VirtualBox Manager window opens (Figure 1-27).

Note 1-27 | Your instructor might have special instructions for the following steps. Check with your instructor before proceeding.

To create a virtual machine (VM) using VirtualBox, complete the following steps:

4. Click **New** in the toolbar and follow the wizard to create a VM, as discussed in the following steps. Give your VM a name. When naming resources like VMs, be sure to think through what information the resource's name should provide, and think about how this name will compare to other resource names when appearing in a list together. For example, you might want to include the VM's OS in the name, such as "Windows11-64bit," or you might want to reference the project in which you created the VM, such as "CapProj1-1." What did you name your VM?
5. Select the ISO image you will use to install an OS on the VM.

Figure 1-27 Use the VirtualBox Manager to create and manage virtual machines



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Note 1-28 | An ISO file (which has the .iso file extension) is a disc image file. It contains all the files and folders of a virtual CD or DVD merged into a single file. The file can be burned to a physical disc, or it can be mounted to a virtual device, such as a VM.

6. If necessary, adjust the selection for the Windows OS so it matches the OS you will install from your ISO file. These steps use Windows 11 Home. Check with your instructor for specific instructions on how to handle the Windows product key. Click **Next**.
7. Change the username and password for the guest OS. Make sure to record this information in your Bitwarden vault for future reference.
8. You can accept all default settings for the VM unless directed otherwise by your instructor. As you go, notice the resources allocated to the VM and respond to the following questions:
   a. How much memory will the VM have?
   b. How much space will the VM's hard disk have?
9. The VM automatically begins to boot. As soon as the message Press any key to boot from CD or DVD appears, press the spacebar so the VM will boot from the ISO file. Your VM starts up and begins the process of installing the operating system. This process should complete on its own.

Note 1-29 | If you have trouble booting to the ISO file, you might need to enable EFI. Note that this setting is typically enabled by default during initial setup. If it was not enabled for your VM, go to the VM's **Settings** window and click **System**. In the Extended Features section, select the check box for **Enable EFI (special OSs only)**.

Also, if the VM struggles to install Windows, consider increasing the VM's available memory in the Settings menu. For example, 64-bit Windows installs more easily with 4 GB of RAM rather than the minimum 2 GB. Keep in mind, though, that any RAM dedicated to a running VM is not available to the host machine.

10. After you have installed Windows in the VM, open the Edge browser. Navigate to a website, such as Google.com, to confirm the VM has a good Internet connection. **Take a screenshot** of your desktop showing your VirtualBox app, your running VM, and the VM's successful connection with the Internet; submit this visual with your responses to this project's questions.
11. When you're finished, be sure to shut down the VM properly—just like a physical machine, a virtual machine can be corrupted by improper shutdowns.
12. Select the VM in the left pane of the VirtualBox Manager window and click **Settings**. In the VM's Settings window, click **Storage** in the left pane.
13. In the Storage Devices area, right-click the "Unattended" image and click Remove Attachment.

In later chapters, you'll continue to build your virtual network and install resources in the VMs on your network. Make sure you record any relevant information, such as user account password or PIN, in your Bitwarden vault.

Solutions to Self-Check Questions

Section 1-1: Network Models

1. Which of the following is part of a network's physical topology?
   Answer: b. A printer plugged into a nearby desktop computer
   Explanation: Physical topology mostly refers to a network's hardware and how computers, other devices (such as **printers**), and cables or radio signals work together to form the physical network. Operating systems, passwords, and file permission settings are all part of a network's logical topology.
2. Which of the following is an advantage of P2P file sharing?
   Answer: d. Inexpensive
   Explanation: Peer-to-peer networks are often **less expensive** to set up and maintain than other types of networks. A network operating system, such as Windows Server, is much more expensive than a desktop operating system, such as Windows 11. P2P networks are not scalable and are not necessarily secure. Client–server networks store user accounts and passwords in one place, and access permissions are centrally granted.
3. What Windows Server group must clients join to access network resources in a client–server network?
   Answer: b. Domain
   Explanation: When Windows Server controls network access to a group of computers, this logical group is called a Windows **domain**. A workgroup is one option for sharing resources in a peer-to-peer network. A server might use a RAID (redundant array of independent disks) configuration of hard drives, so that if one hard drive fails, another hard drive automatically takes its place.

Section 1-2: Network Services

4. Which of the following is a secure protocol?
   Answer: d. SSH
   Explanation: The **SSH (Secure Shell)** protocol creates an encrypted channel or tunnel between two computers, which provides security that Telnet lacks. HTTP (Hypertext Transfer Protocol) requires another protocol, such as TLS (Transport Layer Security), for security. FTP (File Transfer Protocol) does not provide encryption and is, therefore, not secure. However, an encrypted and secure file transfer protocol is SFTP (Secure File Transfer Protocol), which is based on the SSH protocol.

5. What is an example of an open source DBMS?
   Answer: a. MySQL
   Explanation: Popular examples of SQL database software include Microsoft SQL Server, Oracle Database, and the open-source **MySQL**. Open source is the term for software whose code is publicly available for use and modification, which is often considered more secure because users can evaluate the source code of the software to ensure there are no loopholes left open for attackers to exploit. Note that "open source" is not synonymous with "free."

6. Which of these protocols is specific to the Windows OS?
   Answer: c. RDP
   Explanation: In Windows operating systems, **RDP (Remote Desktop Protocol)** provides secure, encrypted transmissions for remotely accessing another computer. A similar service, SSH (Secure Shell), also works in Linux systems to create an encrypted channel or tunnel between two computers. SMTP (Simple Mail Transfer Protocol) is used to send email messages to a mail server. HTTPS (Hypertext Transfer Protocol Secure) uses TLS (Transport Layer Security) to encrypt transmissions between web servers and their clients.

Section 1-3: Network Hardware

7. When you connect your keyboard, mouse, and monitor to your computer, what kind of physical topology are you using?
   Answer: c. Star topology
   Explanation: In a **star topology**, all devices connect to one central device. In this case, the keyboard, mouse, and monitor are connected to the central device, the computer. In a bus topology, the devices would be connected in a line with no device connected to more than two other devices. In a mesh topology, all devices would be connected to each other. In a hybrid topology, two or more other topologies would be used together.

8. Which networking device is best for connecting your network printer to three computers?
   Answer: a. Switch
   Explanation: A **switch** receives incoming data from one of its ports and redirects (or switches) it to another port or multiple ports that will send the data to its intended destination(s). The fundamental difference between a switch and a router is that a switch belongs only to its local network and a router belongs to two or more networks. In this case, all the devices belong to the same network. While a hub would work, it's inefficient and outdated so is not the best choice. A server stores files or applications for other computers to access.

9. Which network types can your smartphone use?
   Answer: e. WAN, LAN, and PAN
   Explanation: When a smartphone communicates over the cellular network for phone calls, it is using a **WAN** across a wide geographical area. When a smartphone connects with your home network, it is using a **LAN** in your home. When a smartphone connects with Bluetooth ear buds, a fitness tracker, or a Bluetooth weight scale, it is using a **PAN** to connect devices intended for use by one person at a time.

Section 1-4: The Seven-Layer OSI Model

10. At what OSI layer does your browser's webpage request begin its journey through the OSI layers?
Answer: a. Application layer
Explanation: The **application layer** in the OSI model describes the interface between two applications, each on separate computers. It takes data from an application or utility and creates a payload for lower OSI layers to manage and transport. The lower layers, such as the network, data link, and physical layers, encapsulate the payload from higher layers to add each protocol's information before passing the message down to the next layer.

11. Which layer holds the address of the target application running on the destination computer?
Answer: b. Transport layer
Explanation: The **transport layer** header addresses the receiving application by a number called a port, which identifies a specific application on a computer. The application layer does not include addressing information. The network layer header identifies the sending and receiving host computers on a network by their IP addresses. The data link frame's header contains the hardware addresses of the source and destination NICs. This address is called a MAC (Media Access Control) address.

12. Which OSI layer adds both a header and a trailer?
Answer: c. Data link layer
Explanation: The **data link layer** puts its control information in a data link layer header and also attaches control information to the end of the packet in a trailer. The entire data link layer message is then called a frame. The transport and network layers both add their information in their respective headers but typically do not use trailers. While there is a preamble added in front of the data link layer header, it's not typically considered to be a physical layer header. The physical layer is responsible for sending bits via a wired or wireless transmission.

Section 1-5: Safety Procedures and Policies

13. Which fire suppression level provides the most targeted protection?
Answer: b. Rack level
Explanation: **Rack level** fire suppression offers targeted fire detection and suppression inside each cabinet that holds network devices to reduce the impact on surrounding equipment. Room level fire suppression incorporates a sprinkler or gas system that provides fire suppression in parts of a room or throughout the entire room. A building level system protects the entire building and employees throughout that space.

14. What is the minimum amount of ESD required to damage sensitive electronic equipment?
Answer: c. 10 volts
Explanation: When your body and a component have different static charges and you touch the component, you can discharge up to 1,500 volts of static electricity without seeing a spark or feeling the discharge. However, it only takes **10 volts** to damage certain components. This means you could fry a motherboard and never even feel a shock.

Section 1-6: Troubleshooting Network Problems

15. What should you do after making a plan to solve a problem and carrying out your plan?
Answer: d. Verify functionality
Explanation: After implementing your solution, you'll test the system to **verify full system functionality**. It's also a good idea to return a few days later and make sure all is working as you expected. Also consider what you can do to make sure the problem doesn't reappear. You find out what has recently changed at the beginning of the process. You test your theory of what's causing the problem before developing a plan to solve the problem. You document your work at the end after verifying the solution worked.

16. While exploring the problem and developing a theory about the cause, where can you go to get information on similar problems your coworkers have faced in the past?
Answer: b. Knowledge base
Explanation: A **knowledge base** is a collection of accumulated insights and solutions to the problems encountered on a particular network. An SDS (safety data sheet) explains how to properly handle substances such as chemical solvents and how to dispose of them. The current troubleshooting ticket in the ticket tracking system will contain information about the current problem but not accumulated insights from previous problems. An instruction manual can provide insights on how a device or application works and possibly troubleshooting help for common problems, but it will not include information on previous problems experienced by your coworkers.